

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5. GENERAL RELATIVITY

TRINITY TERM 2022

Thursday, 16 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$, the Riemann tensor has components

$$R^\gamma_{\alpha\beta\delta} = \frac{\partial}{\partial x^\delta} \Gamma_{\alpha\beta}^\gamma - \frac{\partial}{\partial x^\beta} \Gamma_{\alpha\delta}^\gamma + \Gamma_{\alpha\beta}^\epsilon \Gamma_{\delta\epsilon}^\gamma - \Gamma_{\alpha\delta}^\epsilon \Gamma_{\beta\epsilon}^\gamma.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv -R^\gamma_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2} \frac{\partial^2}{\partial x^\alpha \partial x^\beta} \ln |g| - \frac{\partial \Gamma_{\alpha\beta}^\gamma}{\partial x^\gamma} + \Gamma_{\alpha\gamma}^\delta \Gamma_{\beta\delta}^\gamma - \frac{1}{2} \Gamma_{\alpha\beta}^\gamma \frac{\partial}{\partial x^\gamma} \ln |g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R + \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta} R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy-momentum tensor.

For the line element

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) R + \frac{1}{3} \Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k, \end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. The Riemann curvature tensor can be written as:

$$R_{\lambda\mu\nu\kappa} = \frac{1}{2} \left(\frac{\partial^2 g_{\lambda\nu}}{\partial x^\kappa \partial x^\mu} - \frac{\partial^2 g_{\mu\nu}}{\partial x^\kappa \partial x^\lambda} - \frac{\partial^2 g_{\lambda\kappa}}{\partial x^\nu \partial x^\mu} + \frac{\partial^2 g_{\mu\kappa}}{\partial x^\nu \partial x^\lambda} \right) + g_{\eta\sigma} (\Gamma_{\nu\lambda}^\eta \Gamma_{\mu\kappa}^\sigma - \Gamma_{\kappa\lambda}^\eta \Gamma_{\mu\nu}^\sigma).$$

(a) State its symmetries without proving them.

[4]

(b) It obeys the Bianchi identities that are central to General Relativity, and can be written in this form:

$$R_{\lambda\mu\nu\kappa;\eta} + R_{\lambda\mu\eta\nu;\kappa} + R_{\lambda\mu\kappa\eta;\nu} = 0.$$

What is meant by the “;” symbol? From the Bianchi identities, derive:

$$\left(R^{\mu\nu} - g^{\mu\nu} \frac{R}{2} \right)_{;\mu} = 0.$$

Clearly define all the symbols used and explain why it is useful to construct such a tensor with vanishing divergence.

[7]

(c) Consider the following form of the field equations:

$$R_{\mu\nu} - g_{\mu\nu} \frac{R}{2} = CT_{\mu\nu}.$$

Note the lowered indices. Show that this can be written as:

$$R_{\mu\nu} = CS_{\mu\nu}$$

where $S_{\mu\nu} = T_{\mu\nu} - g_{\mu\nu}T/2$, and C is a constant.

[3]

(d) The Newtonian limit implies a non-relativistic source particle producing a weak and static gravitational field. Write down the definition of the Ricci tensor $R_{\mu\nu}$ and specifically the R_{00} term. At the Newtonian limit, show that $R_{00} = \frac{1}{2}C\rho c^2$, where ρ is the Newtonian mass density and c the speed of light. Write down the Riemann tensor in the weak field limit, and then R_{00} specifically. Simplify R_{00} by assuming the static limit, to show:

$$R_{00} = \frac{1}{2}\nabla^2 g_{00}.$$

[7]

(e) Finally, define C such that Poisson’s equation for gravity is satisfied in the Newtonian limit, taking $g_{00} \simeq -(1 + 2\Phi/c^2)$.

[4]

2. (a) For a curved spacetime, show how the metric tensor $g_{\mu\nu}$ can be expressed in terms of a suitable constant tensor and relations involving a set of locally inertial coordinates ξ^α . [2]

(b) Write down an equation defining the covariant derivative of a vector A^λ , and a clear explanation of its usefulness in a curved spacetime. In your definition, use and define the Levi-Civita affine connection $\Gamma_{\mu\nu}^\lambda$, and state its symmetries. [5]

(c) Using your definitions, show explicitly that the covariant derivative of the metric $g_{\mu\nu}$ vanishes. [4]

(d) Now define the metric tensor using the basis vectors $\mathbf{e}_a$ and $\mathbf{e}_b$. Assuming that the object Γ_{ac}^b is defined through:

$$\frac{\partial \mathbf{e}_a}{\partial x^c} = \Gamma_{ac}^b \mathbf{e}_b,$$

show that:

$$\frac{\partial g_{ab}}{\partial x^c} = \Gamma_{ac}^d g_{db} + \Gamma_{bc}^d g_{ad}.$$

What does this tell you about Γ_{ac}^b ? [4]

(e) For the 2D surface of a 3D sphere of radius R , evaluate $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$ from the transformation laws for the basis vectors, and show that the metric g_{ij} is given by:

$$g_{ij} = R^2 \begin{pmatrix} g_{\theta\theta} & g_{\theta\phi} \\ g_{\phi\theta} & g_{\phi\phi} \end{pmatrix} = R^2 \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix},$$

where θ and ϕ are defined using the standard definition of polar coordinates. [3]

(f) Parallel transport of a vector $\mathbf{V}$ along a curve of constant θ on the surface of this sphere results in the components of $\mathbf{V}$ in θ and ϕ changing according to the equation:

$$V_{;\phi}^a = \frac{\partial V^a}{\partial \phi} + \Gamma_{b\phi}^a V^b = 0.$$

Compute the non-vanishing affine connection coefficients for this surface to show that:

$$V^\theta = A \cos(\phi \cos \theta) + B \sin(\phi \cos \theta)$$

and

$$V^\phi = -A \frac{\sin(\phi \cos \theta)}{\sin \theta} + B \frac{\cos(\phi \cos \theta)}{\sin \theta}.$$

If $\mathbf{V}$ has components $(V^\theta, V^\phi) = (1, 0)$ at $\theta = \pi/3$ and $\phi = 0$, show that the final components of $\mathbf{V}$ when parallel transported along a full circle of ϕ with constant θ , are $(-1, 0)$. Comment on the magnitude of $\mathbf{V}$. Why do you not expect the length of the vector to change? [7]

3. A point source is emitting a beam of photons that is deflected slightly by the presence of a massive body at a large distance to the source.

(a) Using a diagram, describe the path of a single photon. Make sure to carefully indicate the deflection angle δ and the impact parameter b . Explain clearly the reference path with respect to which you are defining the deflection angle. [4]

(b) Assume that spacetime around this massive body follows the Schwarzschild metric. Using a Lagrangian, show that the radial equation of motion can, in general, be written as:

$$\left(\frac{dr}{d\phi}\right)^2 + r^2 B \left(1 + \frac{c^2 r^2}{\gamma_\infty^2 J^2}\right) = \frac{c^2 r^4}{J^2}$$

in standard spherical coordinates, where

$$B = 1 - \frac{2GM}{rc^2}.$$

Explain the meaning of the constants J and γ_∞ . [8]

(c) Show that the path of the photon obeys

$$u'' + u = \frac{3GM}{c^2} u^2$$

where $u = 1/r$ and $'$ indicates derivatives with respect to ϕ . Obtain the solution in the case where the term on the right hand side is zero. Now, by expanding u as a sum of 0th (u_0) and 1st order magnitude (u_1) terms in $1/c^2$, show that, for the first order terms,

$$u_1'' + u_1 = \frac{3GM}{c^2} \frac{(1 - \cos 2\phi)}{2b^2}.$$

[6]

(d) The total deflection can be shown to be $\frac{4GM}{bc^2}$, which you do not need to prove here. A luminous astrophysical point source at a distance D emits light isotropically. For an observer on Earth this source lies on the extended line of sight to a black hole of mass M which is located much nearer to Earth at distance $L < D$. Describe the resultant image of the point source on the sky. Show that the angular size 2θ of the image is given by the equation: [7]

$$2\theta = 2\sqrt{\frac{4GM}{c^2} \frac{(D-L)}{LD}}.$$

4. (a) Assume that the distance to the Moon is 4×10^5 km. If the Earth and the Moon independently drifted with the Hubble flow, then by what velocity would the Moon be receding? Express your answer in cm per year. [3]

(b) The actual recession velocity of the Moon is measured to be 3.8 cm per year. Should modellers of the Earth-Moon system take into account the Hubble expansion of the Universe when they do their calculations? Explain your reasoning very clearly. [5]

(c) Consider a spherical region of radius r_0 with a small excess mass δM (relative to the critical value) in an Einstein-de Sitter (EdS) universe. The initial velocity at the surface r_0 , relative to the centre of the sphere, is exactly what it would be in an EdS universe without the excess mass. Show that the co-moving time τ at which the sphere stops expanding and begins to recontract is given by an equation of the form

$$\int_{r_0}^{A/B} dr \sqrt{\frac{r}{A - Br}} = \tau.$$

Express the constants A and B in terms of the gravitational constant G , M , δM and r_0 , where M is the total mass of the sphere. [6]

(d) Little error is incurred if we set the lower limit of the integral to zero, but retain r_0 everywhere else. Using this approximation, solve for τ in terms of G , M , δM and r_0 . [You are given that $\int_0^1 \sqrt{\frac{u}{1-u}} du = \frac{\pi}{2}$.] [4]

(e) One of the most important tools of modern cosmology is a study of the relative temperature fluctuations, $\Delta T/T$, in the cosmic microwave background (CMB) due to density fluctuations. An over-density causes time to pass a little slower near the fluctuation. Assume that in our above example the fluctuation has been caused by an adiabatic compression. Show that the initial $\Delta T/T$ resulting from time delay is given by the two terms:

$$\frac{\Delta T}{T} = -\frac{G\delta M}{r_0 c^2} - \frac{\Delta R}{R}$$

where $\Delta R/R$ represents the local relative change in the scale factor of the Universe due to the effects of the density fluctuation. Be very clear in your explanation of the physical origin of these two terms. In particular, what is the sign of the final term for an over-density fluctuation? [7]